Normal Operating Display Screens:

INITIALIZATON SCREEN: This Screen is displayed briefly while the unit is powered up.	E-VEHICLE 2.4kW BATTERY CHARGER
STARTUP SCREEN: This Screen is displayed while the charger is starting the charging cycle after detecting the battery.	START BAT:54.5V-00.0A
OPERATION SCREEN1: This screen displays the Battery Bar Graph and Charging Time elapsed in HH:MM in the first line. The Second line shows the actual Battery Voltage and actual Charging current. Please Note, the Battery Bar graph is a tentative indication of battery voltage only and may not accurately represent battery capacity status.	[■■■■] 00:00 BAT:54.3V-40.4A
OPERATION SCREEN2: This screen displays the presently selected Charge Profile, CV voltage and CC Current Settings	TYPE: 57V 40A Li-16s LFP
OPERATION SCREEN3: This screen displays the actual Mains Voltage in the first line, Display Version and Internal Power Module version and Fault status in the second line.	MAINS: 220.0V v0 M1.3 M2.3

Under normal operating conditions, the above last three screens are displayed in rotation continuously.

Fault Messages:

This Screen is displayed when the charger is unable to detect the battery. This usually happens in case of Battery Connector or wiring problem or in case the charging is disabled by the BMS/PCM module of the Battery.	FLT!BAT OPEN OR BMS TRIP 54.5V
This Screen is displayed in case the Battery gets connected in reverse polarity or the charger output is short circuited. This Screen is displayed in in event of Mains voltage out of acceptable range.	FLT! OUTPUT REVERSE OR SHORT
Apart from the above, Fault Message screens are displayed for events like High Temperature, High Battery Voltage, Internal Hardware Fault and Internal Module Communication Fault.	FLT!MAINS

Mfg. & Mktd. by :

ECOSTAR INNOVATION PVT. LTD.

Khasara No-218, Nangli Sakrawati Industrial Area, Najafgarh, New Delhi-110043
E-mail : info@ecostartek.com | Mobile : +91 9818511898

ECO FRIENDLY

1 YEAR WARRANTY

EcostaR®

E-Vehicle
Lithium Battery Charger
2400W

AIS 156 II COMPLIED

DESIGNED & MADE INDIA

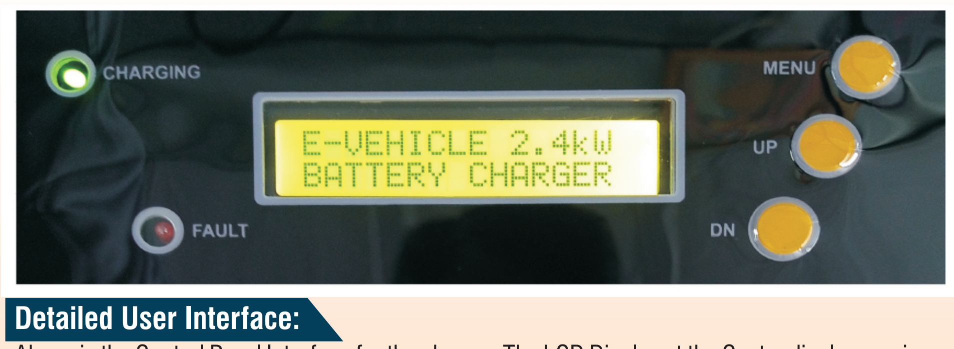
Ecostar 2400W CAN EV Battery Charger is a SMPS type battery charger optimized for charging LFP or NCM or NCA chemistry Li-Ion battery packs for Light E-Vehicle application. The Charger can be user configured using LCD Menu or CAN Communication Interface for charging 13-Cell-Series to 24-Cell-Series Lithium Battery packs.

Standard Features

- Intelligent Four Stage Charging profile for fast and balanced charging of high capacity Li-Ion packs of 60AH - 200AH range.
- Output Voltage Range 50V to 84V. Maximum Output current 40A to 30A. Maximum output power limited to 2400W.
- Features are compliant with the requirements of AIS156 Phase2 amendments.
- Configurable Charge Profile Constant Current and Voltage Parameters through LCD display menu or through CAN Interface.
- User Selectable Pre-Configured charge profile for several common Lithium Battery pack configurations apart from full custom setting mode.
- High Efficiency Charger with Active Power Factor Correction to reduce power consumption.
- High charging current even at low mains voltages for faster charging.
- Informative user interface shows various current and voltage parameters on LCD Display along with LED Indications and audio alarms.
- Electronic Reverse Battery Protection to avoid inconvenient fuse replacement in case of accidental reverse connection.
- Easy replaceable Battery Cable.

Operation Procedure:

Connect the Charger to Battery first using the Battery Connector cable and then apply mains power. The charger will blink both LED indications once and two beep sound at power-up, the Green mains indication will become steady and the charger will start charging. The Battery Bar Indications will start animating with current increasing gradually. As the charging current increases, the charger turns on the internal cooling fan and turns off the fan once the current goes down or internal temperature falls. Once the Battery voltage reaches set voltage and current falls to a pre-defined level, the Battery Bar Indications becomes steady. The battery is almost full charged at this instant and can be disconnected and used for backup. However, the charger continues to charge the battery for some more time till any of the termination conditions is satisfied. Once the battery is fully charged the charger stops and gives a characteristic beep sound with the display showing "CHARGE COMPLETE". The total charging cycle time is also displayed in HH:MM at charge completion.



Detailed User Interface:

Above is the Control Panel Interface for the charger. The LCD Display at the Center displays various parameters and operating conditions. On the Left side of the Display there are two LEDs to indicate charger operating state. On Right side there are three buttons MENU, UP and DOWN. In Normal Operating mode, pressing the MENU button changes the displayed screens – UP and DOWN buttons are inactive. In Configuration Mode, these three buttons can be used for configuring the charging parameters.

Specifications

Max Output Power	2400W MAX	
CV Voltage Range	50V-84V	Configurable through CAN Communication Interface or factory preset .
CC Current Range	15A-40A	
CC Current Tolerance	CC Setting ± 0.5A	
CV Voltage Tolerance	CVSetting ± 0.3V	
Efficiency	90% Typical	
Input Power Factor	Active PF correction, PF>0.98, ITHD < 5%	
Mains Opreating Range	115VAC -280VAC±10V	
Mains Opreating Frequency	40 Hz TO 60 Hz	
Max Input Current	15A (Output current de-rated at low mains to limit max Input current)	
Cooling System	Forced Air Cooled	
LED Indications	2 LEDs-Green-Charging on, Red-Fault.	
LCD Display Parameters	As per normal operating display screens	
Audio Indications (Buzzer)	Charger Startup, Battery Charge Completion – Characteristic Pattern Beep. Mains Failure or any Fault – Single Beeps	
Protection	Reverse Battery (Electronically) Mains High Voltage (Withstand up to 320VAC RMS.), Mains Over Current (by Fuse) Thermal Protection (Electronically)	
Soft Start Rate	Current Rise rate 2.6A/Sec typical. (0 to 40A in 17 Seconds).	
Charge Termination	1. CV Mode Current < 3A for 15Minutes or 2. CV Mode Time Exceeds 1Hr 50Minutes or 3. Total Charging Time Exceeds 7 Hours	
Pre-Charge Setting	If Battery Pack Voltage below 75% of CAN Configured CV Setting, Current Limited to CC/4. If No CAN, below 40V Current Limited to CC/4. Below 33V current limited to 2.0A.	
CAN Interface (Optional)	ISO11898 CAN 2.0B Compliant Physical Interface. Software Protocol customizable as per BMS requirement specification.	
Earth Leakage Protection (Optional)	Inline ELCB Device compliant as per class I of IS 12640-1	
Charger Operating Temprature	0°C TO 45°C	
Humidity	95% RH Non-Condensing	
Battery Cable	2Mtr 6Sqmm with SB75X/SB50 equivalent connector.	
Mains Cable	1.5m, 1.5sqmm 3core ISI Marked Cable With 16A Plug	
Enclosure	Powder coated sheet metal cabinet	
Dimension / Net Weight	350mm x 270mm x 95mm / 6.7kg	